MILITARY SPECIFICATION

FUZE, GUIDED MISSILE, DUMMY

FMU-138(D-1)/B

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification establishes the requirements for manufacture and acceptance of the FMU-138(D-1)/B Dummy Guided Missile Fuze, referred to herein as the dummy fuze.

2. APPLICABLE DOCUMENTS.

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

TT-I-735	Isopropyl Alcohol.

MILITARY

MIL-S-6872	Soldering Process, General Specification for.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department (ESSD), Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

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MIL-T-14757	Trichlorotrifluoroethane.
MIL-C-45662	Calibration System Requirements.
MIL-S-85354	Switch, Inertia, Fuze.

STANDARDS

FEDERAL

FED-STD-102	Preservation, Packaging, and Packing Levels.

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts.
MIL-STD-331	Fuze and Fuze Components, Environmental and Performance Tests for.
MIL-STD-454	Standard General Requirements for Electronic Equipment.
MIL-STD-810	Environmental Test Methods.
MIL-STD-1246	Product Cleanliness Levels and Contamination Control Program.
MIL-STD-1695	Environments, Working, Minimum Standards for.

PUBLICATIONS

Naval Air Systems Command
(Code Ident 30003)

WS-6536	Procedures and Requirements for Preparation and Soldering of Electrical Connections.

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DRAWINGS

Naval Air Systems Command
(Code Ident 30003)

67A87D460	Fusible Link (Dual).
DL 1186AS311	Shipping and Storage Container, Packing and Marking.
DL 1186AS350	Fuze, Guided Missile, Dummy FMU-138(D-1)/B.
SA 498016	Fixture, Vibration.
SA 498019	Fixture, Transportation Vibration.
DL SA2875456	Test Set, Fuze and Air Valve (Tomahawk).
SA 2876062	Fixture, Shock.

(Copies of specifications, standards, publications, and drawings required by contractors in connection with specified procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS (CFR)

49 CFR 171-178	Transportation.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

3. REQUIREMENTS.

NOTE: Drawings and associated documents referenced herein which are not listed in Section 2 are delineated on the DLs cited in Section 2.

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3.1 Item description. The dummy fuze is an electromechanical device that simulates the explosive-loaded FMU-138/B Guided Missile Fuze which maintains an out-of-line explosive train and an electrical firing circuit in the safe (open) position until certain prerequisite events have occurred after missile launch. The dummy fuze contains no explosive elements, and is used in flight test vehicles and for training of personnel in the testing, handling, assembly, and installation of the fuzing system into the warhead of the Tomahawk Cruise Missile. The general configuration of the dummy fuze is shown on Figure 1.

3.1.1 Government-furnished property.

3.1.1.1 Autonomous components. When specified in the contract or purchase order, the Government will furnish the contractor with the following autonomous components for each dummy fuze produced under a specified contract (see 6.2.1):

a. One fuze inertia switch (MIL-S-85354).

b. One fusible link, dual (Drawing 67A87D460).

3.1.1.2 Support equipment. When specified in the contract or purchase order, the Government will furnish the independent test facility designated in the contract or purchase order with the following support equipment to perform the environmental tests specified herein (see 6.2.1):

a. One vibration fixture (Drawing SA 498016).

b. One transportation vibration fixture (Drawing SA 498019).

c. One shock fixture (Drawing SA 2876062).

3.1.2 Government-loaned property. When specified in the contract or purchase order, the Government will loan the contractor one Tomahawk fuze and air valve test set (DL SA2875456), referred to herein as the test set (see 6.2.1).

3.2 Characteristics.

3.2.1 Performance.

3.2.1.1 Electrical. The requirements of 3.2.1.1.1 and 3.2.1.1.2 shall be met by the dummy fuze during and after assembly in the configurations shown in Table I. In Tables II, III, and IV, test modes and switch position numbers refer to the test set (see 3.1.2), and correspond to the pin numbers in the dummy fuze connector. An electrical schematic for the dummy fuze is shown on Figure 2.

3.2.1.1.1 Resistance (safe position). The dummy fuze shall meet the resistance requirements of Table II for each of the three configurations (A, B, and C) specified in Table I while in the safe position (see 6.3.1).

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TABLE I. Assembly configurations for electrical and mechanical operability requirements.

Configuration	Welded[1/]	Sealed[1/]
A	No	No
B	Yes	No
C	Yes	Yes

[1/] See Drawing 1186AS350.

3.2.1.1.2 Resistance (armed position). The dummy fuze shall meet the resistance requirements of Table III for configuration A, Table I, while in the armed position (see 6.3.2). The dummy fuze shall meet the resistance requirements of Table IV for configurations B and C, Table I, while in the armed position (see 6.3.2).

3.2.1.2 Mechanical operability. The dummy fuze shall meet the mechanical operability requirements of Table V for configurations A, B, and C, Table I. Beginning and ending times are indicated by the beginning and ending switching functions. (The requirements of Table V represent dummy fuze calibration times and not dummy fuze arming times.)

3.2.1.3 Seal. At the contractor's facility, the leakage rate of the dummy fuze shall not exceed 5×10^{-5} standard atmosphere cubic centimeters per second (std atm cm^3/s) when measured within 4 hours after pressurizing and sealing. Anytime after shipment from the contractor's facility, the leakage rate of the dummy fuze shall not exceed 5×10^{-4} std atm cm^3/s.

3.2.2 Environmental conditions. The dummy fuze shall not be damaged nor shall performance be degraded by exposure to the following environments.

3.2.2.1 Transportation vibration-temperature. The dummy fuze shall meet the requirements of MIL-STD-331, Test 119, Procedure 2, and shall be operable (see 6.3.3) after exposure to the environment of 4.4.4.1.

3.2.2.2 Shock. The dummy fuze shall meet the requirements of MIL-STD-202, Method 213, Test Condition J, except the amplitude shall be 15 ±1 gravity units (g). The dummy fuze shall be operable (see 6.3.3) after exposure to the environment of 4.4.4.2.

3.2.2.3 Temperature and humidity. The dummy fuze shall meet the requirements of MIL-STD-331, Test 105, either method, and shall be operable (see 6.3.3) after exposure to the environment of 4.4.4.3.

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TABLE II. Resistance (safe position) requirements for configurations A, B, and C.

Test mode	Ohmmeter scale reading	Switch positions		Resistance				Items checked (see Figure 2)
				Configuration				
				During manufacture		After shipment		
				A, B, and C		C		
HP	Off	HP	C-	Lower limit	Upper limit	Lower limit	Upper limit	
HP1	---	(1-8, 10-12)	9	100 Mohms	∞	100 Mohms	∞	HP
HP2	---	(1-12)	13 1/	100 Mohms	∞	100 Mohms	∞	HP
3469	On	C+	C-	Lower limit	Upper limit	Lower limit	Upper limit	(see Figure 2)
S1	10 Mohms	5	8	2/	2/	2/	2/	C1, R1, 3
S2	100 kohms	All	9	Open	Open	Open	Open	---
S3	100 kohms	7	3	Open	Open	Open	Open	S1E
S4	100 kohms	4	12	Open	Open	Open	Open	Inertia switch
S5	100 kohms	4	1	Open	Open	Open	Open	CR6
S6	100 kohms	1	4	35 kohms	45 kohms	20 kohms	60 kohms	CR6
S7	100 kohms	8	10	27 kohms	38 kohms	25 kohms	41 kohms	R4, 6, 8
S8	100 kohms	11	8	26 kohms	34 kohms	10 kohms	34 kohms	$Q1_E$, R4, 6, 7, 8
S9	1 kohm	6	11	295 ohms	365 ohms	260 ohms	400 ohms	R7, $Q1_C$
S10	100 ohms	11	10	71 ohms	95 ohms	65 ohms	115 ohms	$Q1_C$, R7
S11	1 ohm	4	8	---	---	---	---	---
S11 (Cal) 3/	1 ohm	4	8	---	---	---	---	---
S11 (Net) 4/	---	---	---	0.40 ohm	0.90 ohm	0.30 ohm	1.00 ohm	F2
S12	1 ohm	6	10	---	---	---	---	---
S12 (Cal) 3/	1 ohm	6	10	---	---	---	---	---
S12 (Net) 4/	---	---	---	0 ohm	0.45 ohm	0 ohm	1.00 ohm	S1F

1/ Clip must be attached to case for these tests.

2/ Lower limit will show capacitor (C1) charging. Upper limit (approximately 7 Megohms (Mohms)) may be exceeded, depending on charging time.

3/ Calibration readings are taken at calibration socket.

4/ Net values are the normal readings minus the calibration readings.

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TABLE III. Resistance (armed condition) requirements for configuration A.

Test mode 3469	Ohmmeter scale reading	Switch positions		Resistance		Items checked (see Figure 2)
		C+	C-	Lower limit	Upper limit	
A1	100 kohms	10	5	35 kohms	45 kohms	CR5, S1D
A2	100 kohms	5	10	Open	Open	CR5
A3	1 ohm	7	3	---	---	---
A3 (Cal)1/	1 ohm	7	3	---	---	---
A3 (Net)2/	---	---	---	0 ohm	0.45 ohm	S1E

1/ Calibration readings are taken at calibration socket.
2/ Net values are the normal readings minus the calibration readings.

TABLE IV. Resistance (armed position) requirements for configurations B and C.

Test mode 3469	Ohmmeter scale reading	Switch positions		Resistance				Items checked (see Figure 2)
				Configuration				
				B		C		
		C+	C-	Lower limit	Upper limit	Lower limit	Upper limit	
A1	100 kohms	10	5	35 kohms	45 kohms	20 kohms	60 kohms	CR5, S1D
A2	100 kohms	5	10	Open	Open	Open	Open	CR5
A3	1 kohm	8	12	500 ohms	700 ohms	500 ohms	700 ohms	R10, S1B, CR3, S1C
A4	1 kohm	12	8	500 ohms	700 ohms	500 ohms	700 ohms	R9, S1A, CR4, F1
A5	1 ohm	7	3	---	---	---	---	---
A5 (Cal)1/	1 ohm	7	3	---	---	---	---	---
A5 (Net)2/	---	---	---	0 ohm	0.45 ohm	0 ohm	1.00 ohm	S1E

1/ Calibration readings are taken at calibration socket.
2/ Net values are the normal readings minus the calibration readings.

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TABLE V. Mechanical operability requirements for configurations A, B, and C.

Test	Pressure applied	Solenoid Vdc		Requirements	
		Configuration			
		A and B (Vdc)	C (Vdc)	A and B	C
All go	33.0 ±0.5 inches water	22 ±1	---	Arm[1/]	Arm[1/]
All go	40.0 ±0.5 inches water	---	22 ±1	Arm[1/]	Arm[1/]
No go	25.0 ±0.5 inches water	24 ±1	28 ±1	Remain safe	Remain safe
Timing	2.0 ±0.1 lb/in^2	24 ±1	28 ±1	11.0 ±1.5 seconds[2/]	11.75 ±2.25 seconds[2/]
Burst	10.0 ±0.3 lb/in^2	24 ±1	28 ±1	Arm[1/]	Arm[1/]

[1/] Rotor (Drawing 642AS5584) fully rotated and locked.

[2/] Rotor fully rotated and locked with time measured as shown on Figure 3.

3.2.2.4 Captive flight vibration (safe position). While in the safe position (see 6.3.1), the dummy fuze shall meet the requirements of MIL-STD-810, Method 514, Procedure V, Part 2, Curve AE, except random vibration shall be applied for 2 hours minimum along each of the three mutually perpendicular axes for a total of 6 hours. The dummy fuze shall be operable (see 6.3.3) after exposure to the environment of 4.4.4.4.

3.2.2.5 Missile flight vibration (armed position). While in the armed position (see 6.3.2), the dummy fuze shall meet the requirements of MIL-STD-810, Method 514, Procedure V, Part 2, spectral density of 0.05 gravity unit squared per Hertz (g^2/Hz), except random vibration shall be applied for 10 minutes minimum along each of the three mutually perpendicular axes for a total of 30 minutes. The dummy fuze shall remain in the armed position during and shall be operable (see 6.3.3) after exposure to the environment of 4.4.4.5.

3.2.3 Visual. The dummy fuze shall meet the following requirements:

a. There shall be no evidence of external damage, such as cracks, broken windows, dents, or other blemishes, which may interfere with functioning or fit at the next assembly.

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b. The rotor (Drawing 642AS5584) shall be in the safe position (see 6.3.1), as evidenced by a white letter "S" on a green background centered in the window located on the dummy fuze housing.

3.3 First article (preproduction sample). Unless otherwise specified in the contract or purchase order, the contractor shall furnish a preproduction sample of 30 dummy fuzes to the testing activity designated in the contract or purchase order for the first article inspection and approval (see 4.1.2.1 and 6.2.1). The preproduction sample shall be manufactured using the same methods, materials, processes, and procedures proposed for production. Any production prior to acceptance of the preproduction sample shall be at the risk of the contractor.

3.4 Construction. The construction of the dummy fuze shall meet the requirements specified herein and on the drawings and documents listed on DL 1186AS350.

3.4.1 Production drawings. The dummy fuze shall meet the requirements of and be manufactured and assembled in accordance with the drawings and documents listed on DL 1186AS350.

3.4.2 Standards of manufacture.

3.4.2.1 Product cleanliness. Product cleanliness shall meet the requirements of MIL-STD-1246, Level 200. No oils or solid lubricants of any kind shall be used on any part of the dummy fuze after final cleaning of components, except as specified on the drawings or herein.

3.4.2.2 Working environment. The working environment shall meet the minimum levels and requirements of MIL-STD-1695.

3.4.3 Test equipment calibration. When specified in the contract or purchase order, test equipment shall be calibrated and maintained in accordance with a plan meeting the requirements of MIL-C-45662. The calibration plan shall be prepared in accordance with the Data Item Description (DID) cited in 6.2.2 and shall be approved by the Government prior to use.

3.4.4 Fuze inspection record. When specified in the contract or purchase order, an inspection record, as shown on Figure 4 and prepared in accordance with the DID cited in 6.2.2, shall be properly filled in for each dummy fuze produced under a specified contract. The required test data shall be extracted from the inspection and acceptance records maintained by the contractor.

3.4.5 Workmanship. The dummy fuze, including all parts and accessories, shall be manufactured in a thoroughly workmanlike manner in

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accordance with the applicable drawings and documents specified herein and on DL 1186AS350. Workmanship shall meet the requirements of MIL-STD-454, Requirement 9.

3.4.5.1 Soldering. Preparation, procedures, and requirements for soldering of electrical connections shall be in accordance with WS-6536. Mechanical soldering shall be in accordance with MIL-S-6872.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein (see 6.2.1). Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government (see 6.2.1). The Government reserves the right to perform or witness any of the inspections set forth in this specification where such inspections are deemed necessary to ensure that supplies and services conform to prescribed requirements.

4.1.1 Classification of inspections. Inspection requirements specified herein are classified as follows:

a. Preproduction sample inspections (see 4.1.2.1).

b. Quality conformance inspections (see 4.2).

1. Individual unit acceptance inspections (see 4.2.3).

4.1.2 Special tests and examinations.

4.1.2.1 Preproduction sample inspections. The preproduction sample of 30 dummy fuzes shall be subjected to the individual unit acceptance inspections of Figure 5 (see 4.2.3). The preproduction sample shall then be subjected to the inspections of Figure 6 in the sequence shown. Inspections below the dashed line of Figure 6 shall be performed at an independent facility designated in the contract or purchase order (see 6.2.1).

4.1.2.1.1 Preproduction sample acceptance and rejection criteria. When subjected to the inspections of Figure 6, each dummy fuze of the preproduction sample shall meet the applicable performance requirements of

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Section 3 on the first and only trial. Defects are classified as CRITICAL or MAJOR, as defined in MIL-STD-105, and are specified on the drawings. The preproduction sample shall be rejected if:

a. One or more CRITICAL defects are found, or

b. Two or more MAJOR defects are found.

NOTE: Although one MAJOR defect does not cause rejection of the preproduction sample, appropriate corrective action shall be taken prior to production to prevent recurrence.

4.2 Quality conformance inspections.

NOTE: The contractor may, prior to forming and submitting a lot for acceptance, subject the dummy fuze to the inspections of 4.4, except for 4.4.4, to the extent deemed necessary.

4.2.1 Inspection lot. Unless otherwise specified in the contract or purchase order, an inspection lot shall contain not less than 51 nor more than 280 dummy fuzes, including those required for destructive tests (see 6.2.1). An inspection lot, as defined in MIL-STD-105, shall consist of completed dummy fuzes fabricated under the same conditions and offered for inspection at the same time.

4.2.1.1 Serial numbers. Unless otherwise specified in the contract or purchase order, all dummy fuzes shall be assigned a Government serial number in order of consecutive assembly (see 6.2.1). After assignment of a Government serial number to each dummy fuze, the inspection lot shall be formed and identified by lot number.

4.2.2 Quality conformance inspection sequence. Prior to formation of the lot, the dummy fuze shall be subjected to the individual unit acceptance inspections of Figure 5 (see 4.2.3). After the inspection lot has been formed and identified by number, the lot shall be subjected to the inspections above the dashed line of Figure 7. Fifteen dummy fuzes shall then be randomly sampled from the inspection lot in accordance with MIL-STD-105 and subjected to the inspections below the dashed line of Figure 7 at an independent facility designated in the contract or purchase order (see 6.2.1). The remainder of the lot shall be held in bond pending acceptance of the 15-dummy fuze sample.

4.2.2.1 Quality conformance acceptance and rejection criteria. Defects in the dummy fuze are classified as CRITICAL or MAJOR, as defined in

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MIL-STD-105, and are specified on the drawings. Acceptance and rejection criteria shall be based on the following:

a. For the inspections of Figure 5, all parts, subassemblies and assemblies found to be defective shall be rejected on an individual basis and shall not be used in further assembly of the dummy fuze (see 4.2.3).

b. For the inspections above the dashed line of Figure 7:

1. Where 100 percent inspection is required, all defective dummy fuzes shall be rejected.

2. Where sampling is required for lot sizes of 51 to 159 dummy fuzes, lot acceptability shall be determined in accordance with MIL-STD-105, Single Sampling Plan, Tightened Inspection, sample size code letter H, AQL of 0.40 percent defective. For lot sizes of 160 to 280 dummy fuzes, lot acceptability shall be determined in accordance with MIL-STD-105, Double Sampling Plan, Normal Inspection, sample size code letter K, AQL of 0.04 percent defective.

c. For the inspections below the dashed line of Figure 7, the failure of any sample dummy fuze to meet the applicable requirements of Section 3 shall cause rejection of the lot.

4.2.2.1.1 Counting of defects. When counting defects found in dummy fuzes to determine acceptability of the lot, dummy fuzes with multiple defects shall be counted as a single defect, using the highest classification of the defects found. For example, when a dummy fuze has both a CRITICAL and MAJOR defect, it shall be counted as a CRITICAL defect.

4.2.3 Individual unit acceptance inspections. Individual unit acceptance inspections shall be performed on all parts, subassemblies, and assemblies during the manufacture of the dummy fuze in accordance with Figure 5. Parts, subassemblies, or assemblies found to be defective during the inspections shall not be used in further assembly of the dummy fuze.

4.3 Test equipment and conditions.

4.3.1 Standard test equipment. The contractor shall provide and maintain an adequate system of inspection and test equipment necessary to ensure that parts and products will meet the contract or purchase order, specification, and drawing requirements. Unless otherwise specified herein, the magnitude of any error introduced by the test equipment shall not exceed 10 percent of the allowable tolerance of the requirement being measured. All test equipment shall be calibrated as specified in 3.4.3.

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4.3.2 Special test equipment and circuits. When special test equipment and circuits, other than as specified herein or loaned by the Government (see 3.1.2), are devised or commercially-available equipment is employed, all test equipment, circuits, and methods shall meet the requirements of 4.3.1 and are subject to approval by the procuring activity.

4.3.3 Gages. The contractor shall provide whatever gages are necessary to ensure conformance with the dimensional requirements of this specification and the applicable drawings. When specified in the contract or purchase order, the procuring activity will furnish drawings of pertinent Government final inspection gages (see 6.2.1). This procedure does not relieve the contractor of his responsibility for the design and manufacture of the gages required for satisfactory fulfillment of the contract requirements, but is intended to facilitate acceptance of all components and assemblies by Government final acceptance gages.

4.3.4 Test environment. Unless otherwise specified herein, the inspection methods of 4.4 shall be performed under the conditions specified in 3.4.2.2.

4.3.5 Resetting. Before detachment from the test set, all dummy fuzes shall be reset to the safe position (see 6.3.1), using the test set, after any test where arming occurs. Although failure of a dummy fuze to reset to the safe position does not affect acceptance of a lot, any dummy fuze which cannot be reset shall be rejected.

4.4 Methods of inspection.

> NOTE: Unless otherwise specified, verify that the dummy fuze is in the safe position (see 6.3.1) before removing the dummy fuze from the test set at any time during any test sequence. Discharge the firing capacitor by shorting pins 5 and 8 together for 15 seconds minimum. Verify that the firing capacitor has been discharged by measuring the voltage across pins 5 and 8; if less than 1 Vdc, the firing capacitor is discharged.

4.4.1 Electrical. The tests of 4.4.1.1 and 4.4.1.2 shall be performed on dummy fuzes during and after assembly as specified on the applicable figures. Resistance measurements shall be performed using the test set (see 3.1.2 and 3.2.1.1), or standard test equipment that meets the requirements of 4.3.1. The test results shall be recorded on the inspection record (see Figure 4).

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4.4.1.1 Resistance (safe position). To determine conformance to 3.2.1.1.1, verify the resistance requirements of Table II with the dummy fuze in the safe position (see 6.3.1). The high pot (HP) tests shall be performed at 500 Vdc ±10 percent. The resistance shall be within the limits specified in the table.

4.4.1.2 Resistance (armed position). To determine conformance to 3.2.1.1.2, verify the resistance requirements of Tables III and IV with the dummy fuze in the armed position (see 6.3.2). The tests for configuration C (see Table I) of Table IV shall be performed during the environmental inspection sequences below the dashed lines of Figures 6 and 7. The resistance shall be within the limits specified in the tables.

4.4.2 Mechanical operability. To determine conformance to 3.2.1.2, verify the mechanical operability requirements of Table V using the test set (see 3.1.2 and 3.2.1.1), or standard test equipment with a two-channel (minimum) recorder and an acoustic pickup which meets the requirements of 4.3.1. Timing shall be determined from the output record chart as shown on Figure 3. Record the test results on the inspection record (see Figure 4).

NOTE: An acoustic pickup shall be used as an audio aid to verify dummy fuze arming and resetting (see 4.3.5).

4.4.3 Seal. To determine conformance to 3.2.1.3, test the dummy fuze in accordance with MIL-STD-202, Method 112, Test Condition C, Procedure IV. Trichlorotrifluoroethane (MIL-T-14757), or isopropyl alcohol (TT-I-735, grade optional) may be used as alternative immersion fluids for gross-leak testing. At the contractor's facility, the dummy fuze shall be tested within 4 hours after pressurizing and sealing and the test results shall be recorded on the inspection record (see Figure 4).

4.4.4 Environmental. All dummy fuzes of the preproduction sample, or a randomly selected sample of 15 fuzes from an inspection lot shall be subjected to the following environmental tests in accordance with Figure 6 or 7, as applicable, at an independent test facility designated in the contract or purchase order (see 6.2.1). All dummy fuzes subjected to these tests shall be considered destroyed and shall not be returned to the lot nor forwarded for service use, but shall be labeled appropriately and disposed of at the discretion of the contracting officer.

4.4.4.1 Transportation vibration-temperature. To determine conformance to 3.2.2.1, mount the dummy fuze to the vibration fixture (Drawing SA 498019). Test the dummy fuze in accordance with MIL-STD-331, Test 119, Procedure 2.

NOTE: For the vibration and shock tests specified herein, the test fixtures shall be mounted to a vibration table or shock machine and the mounting bolts shall be tightened to a torque of 25 ±1 foot-pounds.

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4.4.4.2 Shock. To determine conformance to 3.2.2.2, mount the dummy fuze to the shock fixture (Drawing SA 2876062; alternate, vibration fixture of Drawing SA 498016). Test the dummy fuze in accordance with MIL-STD-202, Method 213, Test Condition J, except the amplitude shall be 15 ±1 g.

4.4.4.3 Temperature and humidity. To determine conformance to 3.2.2.3, test the dummy fuze in accordance with MIL-STD-331, Test 105, either method. The connector shall be thoroughly cleaned and dried after the test.

4.4.4.4 Captive flight vibration (safe position). To determine conformance to 3.2.2.4, mount the dummy fuze to the vibration fixture (Drawing SA 498016). Test the dummy fuze in accordance with MIL-STD-810, Method 514, Procedure V, Part 2, Curve AE, except apply random vibration for 2 hours minimum along each of the three mutually perpendicular axes for a total of 6 hours.

4.4.4.5 Missile flight vibration (armed position). To determine conformance to 3.2.2.5, mount the dummy fuze to the vibration fixture (Drawing SA 498016) while in the armed position (see 6.3.2). Test the dummy fuze in accordance with MIL-STD-810, Method 514, Procedure V, Part 2, spectral density of 0.05 g^2/Hz. Apply random vibration for 10 minutes minimum along each of the three mutually perpendicular axes for a total of 30 minutes. Throughout the test, electrically monitor the dummy fuze to determine whether it remains in the armed position (see 6.3.2).

4.4.5 Visual inspection. Visually inspect the dummy fuze to determine conformance to 3.2.3 and the general requirements of Section 3.

4.4.6 Drawing inspections. To determine conformance to 3.4.1, perform drawing inspections on each lot of parts, subassemblies, and assemblies of the dummy fuze during manufacture, and on the completely assembled dummy fuze. The sampling plans and acceptance criteria shall be as specified on the drawings.

4.4.7 Packaging inspection. Prior to shipment, inspect the dummy fuze to determine conformance to the requirements of Section 5.

5. PACKAGING.

5.1 Preservation-packaging. Preservation and packaging shall be in accordance with FED-STD-102, Level C, and as specified on the drawings listed on DL 1186AS311.

5.1.1 Level C. Dummy fuzes shall be packaged in accordance with the drawings listed on DL 1186AS311 and in a manner that will afford

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adequate protection against corrosion, deterioration, and physical damage during direct shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be in accordance with FED-STD-102, Level C, and as specified on the drawings listed on DL 1186AS311.

5.2.1 Level C. Dummy fuzes, packaged in accordance with 5.1.1, shall be packed in shipping containers as specified on the drawings listed on DL 1186AS311 and in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. Packing and shipping containers shall conform to 49 CFR 171-178.

5.3 Marking. In addition to any special marking required by the contract or purchase order, all marking shall be in accordance with the drawings listed on DL 1186AS311 and MIL-STD-129 (see 6.2.1).

6. NOTES.

6.1 Intended use. The dummy fuze is intended for use in flight test vehicles and during training of personnel in the testing, handling, assembly, and installation of the fuzing system into the warhead of the Tomahawk Cruise Missile.

6.2 Ordering data.

6.2.1 Procurement requirements. Procurement documents should specify the following:

a. Number, title, and date of this specification.

b. Government-furnished property. The contracting officer should arrange to furnish the property listed in 3.1.1.

c. Government-loaned property. The contracting officer should arrange to loan the property cited in 3.1.2.

d. First article. When a first article is required, it should be tested and approved under the appropriate provisions of 7-104.55 of the Defense Acquisition Regulations (DAR). The first article should be a preproduction sample consisting of 30 dummy fuzes as specified in 3.3 and 4.1.2.1. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for examinations, tests, and approval of the first article.

e. Responsibility for inspections, if other than as specified in 4.1.

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f. Inspection facility, if other than as specified in 4.1.

g. Independent facility to perform the inspections below the dashed lines of Figures 6 and 7 (see 4.1.2.1, 4.2.2, and 4.4.4).

h. Inspection lot size, if other than as specified in 4.2.1.

i. Assignment of serial numbers, if other than as specified in 4.2.1.1.

j. Government final inspection gage drawings (see 4.3.3).

k. Special marking, if required (see 5.3).

6.2.2 Data requirements. When this specification is used in a procurement which incorporates a Contract Data Requirements List (DD Form 1423) and invokes the provisions of 7-104.9(n) of the DAR, the data requirements identified below will be developed as specified by an approved DID (DD Form 1664) and delivered in accordance with the approved DD Form 1423 incorporated in the contract. When the provisions of DAR 7-104.9(n) are not invoked, the data specified below will be delivered by the contractor in accordance with the contract requirements. Deliverable data required by this specification is cited as follows:

Paragraph	Data Requirement	Applicable DID
3.4.3	Calibration Plan	UDI-T-23742A
3.4.4	Inspection Record	UDI-M-20404

(Copies of DIDs required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

6.3 Definitions.

6.3.1 Safe position. Visually, the dummy fuze is considered to be in the safe position when a white letter "S" on a green background is centered in the viewing window located in the housing. Electrically, the dummy fuze is considered to be in the safe position when, as measured on the test set, the limits of Table II are met under the conditions specified in the table.

6.3.2 Armed position. Visually, the dummy fuze is considered to be in the armed position when a white letter "A" on a red background is centered in the viewing window located in the housing. Electrically, the dummy fuze is considered to be in the armed position when, as measured on the test set, the limits of Tables III and IV are met under the conditions specified in the tables.

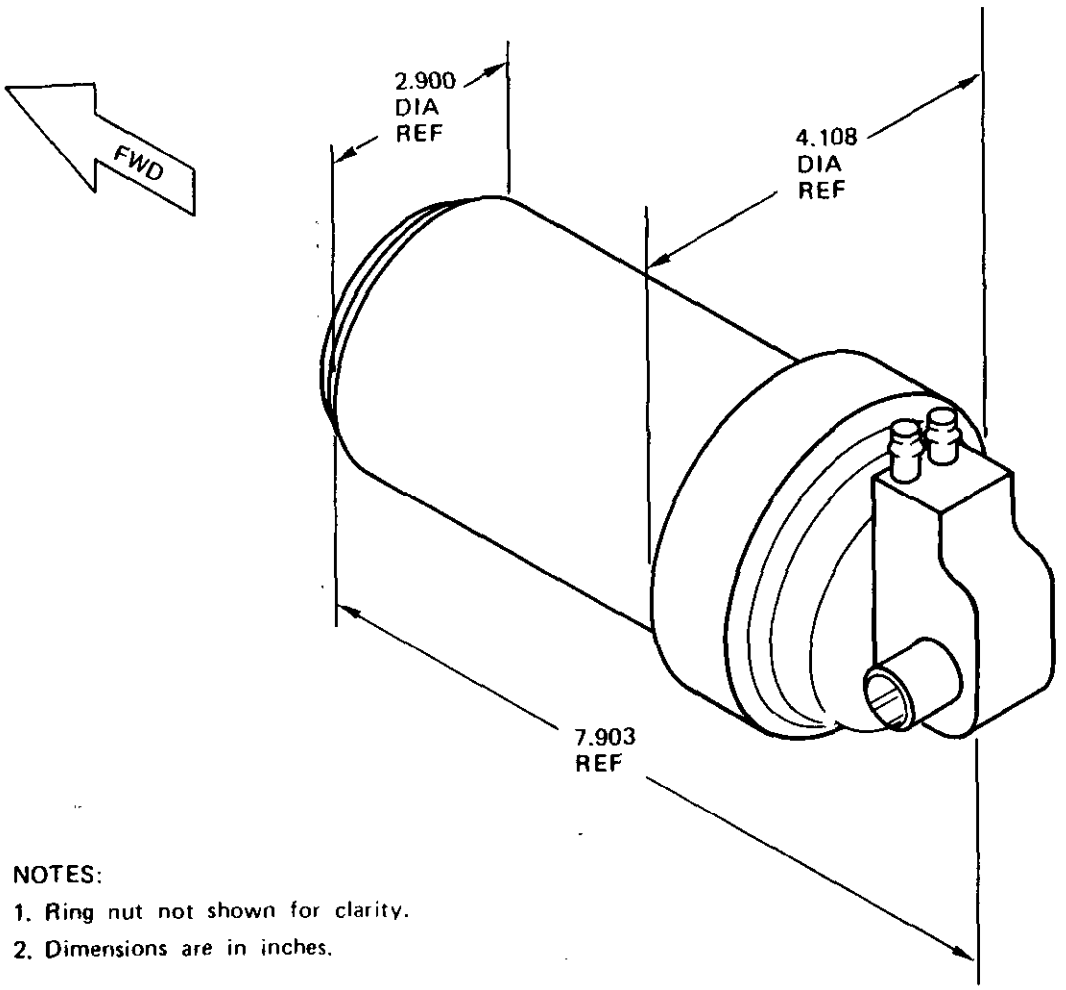
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6.3.3 Operable. Operable is defined as the ability of the dummy fuze, with specified inputs, to perform to completion its specified function and sequence, and produce the specified outputs within the required operational period or at the specified point of time.

Preparing activity:
Navy - AS

(Project 1336-N365)

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NOTES:
1. Ring nut not shown for clarity.
2. Dimensions are in inches.

FIGURE 1. FMU-138(D-1)/B Dummy Guided Missile Fuze general configuration.

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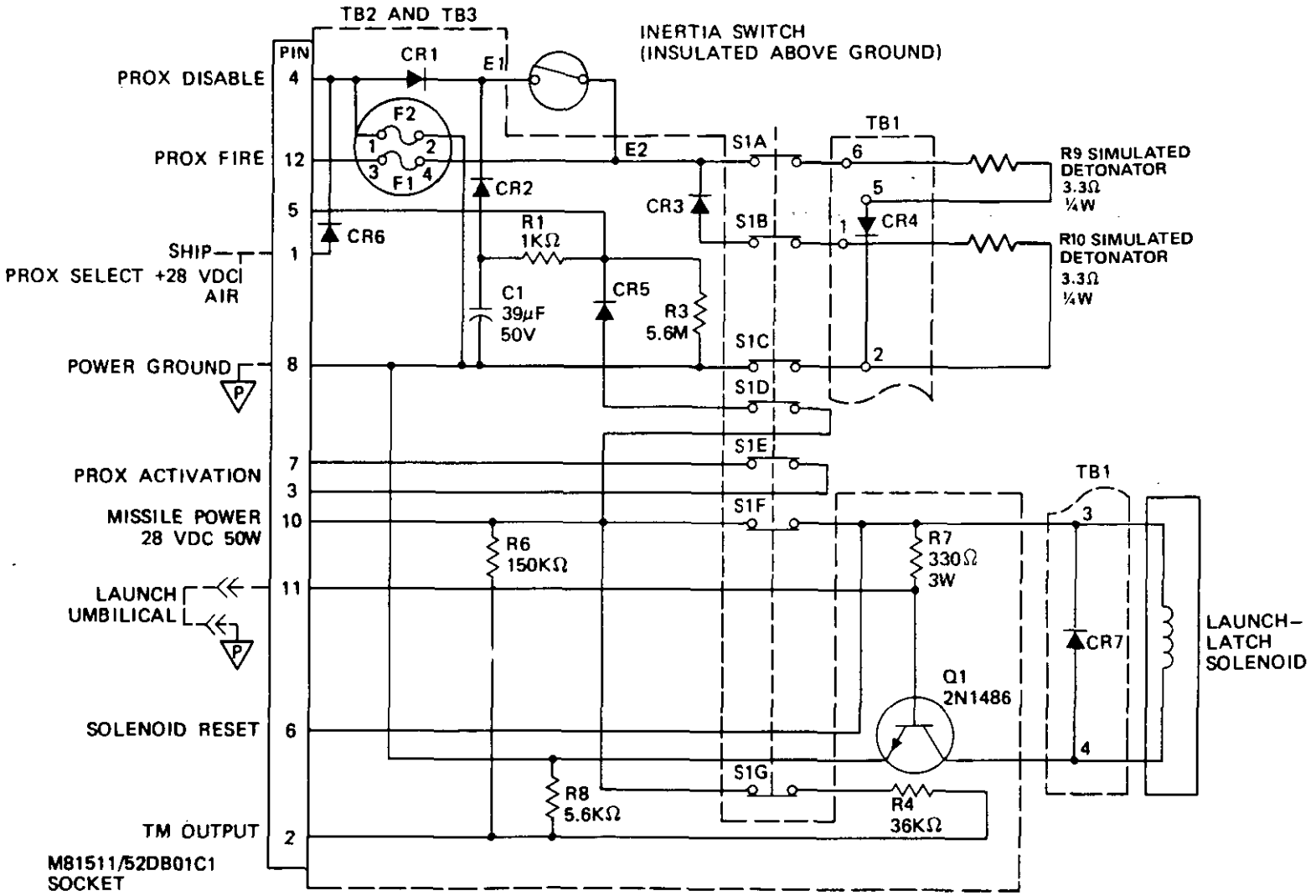


FIGURE 2. Electrical schematic.

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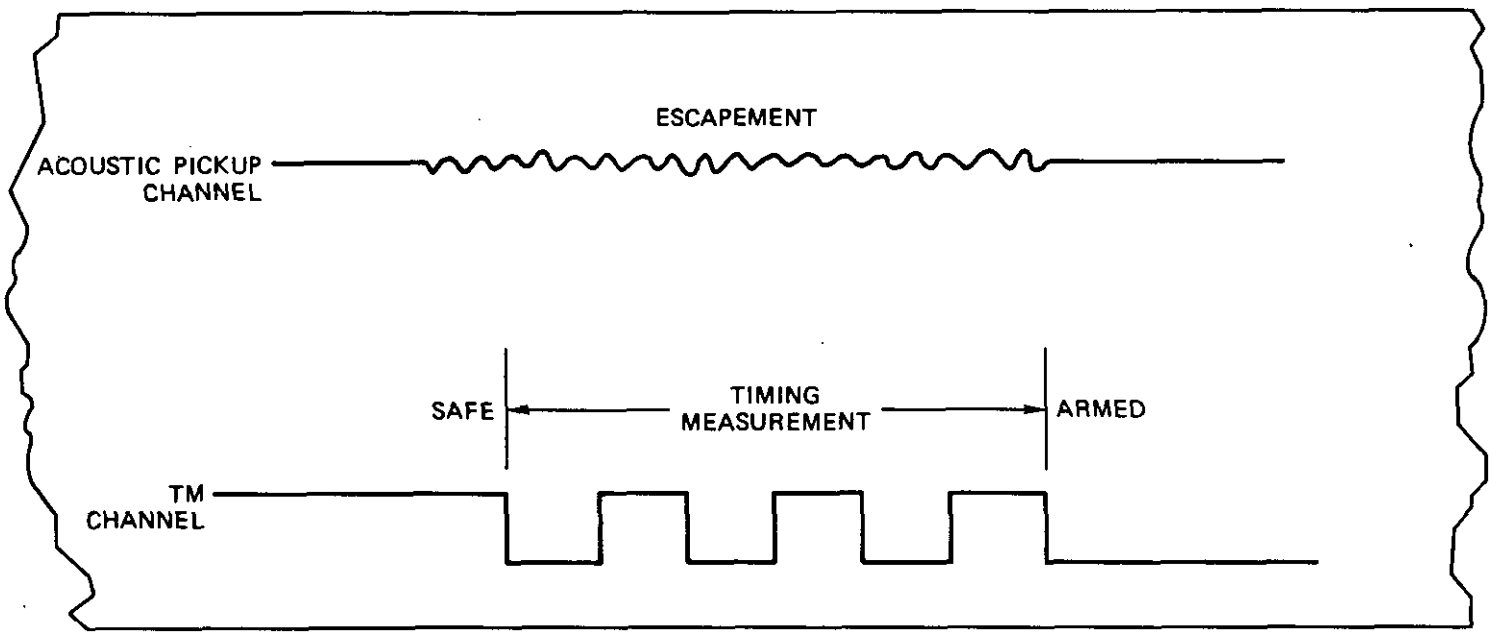


FIGURE 3. Sample of test set dummy fuze timing output record chart.

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INSPECTION RECORD
FMU-138(D-1)/B DUMMY GUIDED MISSILE FUZE

CONTRACT NO. _____ MANUFACTURER _____

PREPRODUCTION SAMPLE ☐
INSPECTION LOT ☐
REWORK ☐

S/N _____ LOT NO. _____

INSPECTOR _____ DATE _____

ELECTRICAL RESISTANCE

SAFE CONDITION								ARMED CONDITION		
TEST	CONFIG A	CONFIG B	CONFIG C	TEST	CONFIG A	CONFIG B	CONFIG C	TEST	CONFIG A	CONFIG B
HP1				S6				A1		
HP2				S7				A2		
S1				S8				A3		
S2				S9				A4	NA	
S3				S10				A5		
S4				S11				CHECKED BY _____		
S5				S12				DATE _____		

DRAWING INSPECTIONS

DWG 1186AS217 VERIFY NOTE 12

TEST	OHMS	TEST	OHMS
13		23	
14		26	
22			

CHECKED BY _____ DATE _____

PARTS/SUBASSEMBLIES
VERIFIED BY _____ DATE _____

DWG 1186AS350 – FINAL ASSEMBLY
VERIFIED BY _____ DATE _____

MECHANICAL OPERABILITY

TEST	RESULT	TEST	RESULT
ALL GO		TIMING	
NO GO		BURST	

CHECKED BY _____ DATE _____

SEAL

LEAK RATE _____ STD ATM CM^3/S

CHECKED BY _____ DATE _____

VISUAL INSPECTION

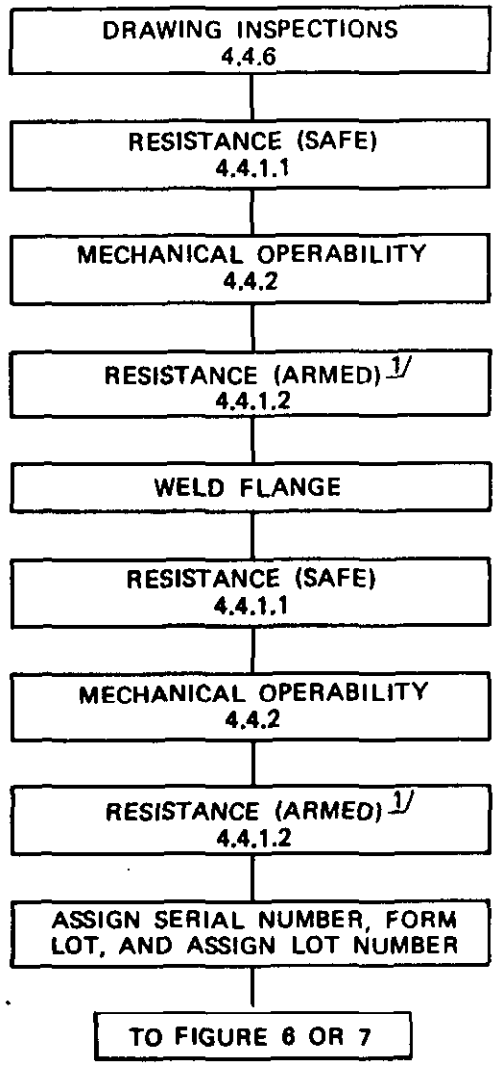
	YES	NO
(1) EXTERNAL DAMAGE	☐	☐
(2) FUZE IN SAFE POSITION (WHITE "S" VISIBLE THROUGH WINDOW)	☐	☐

GOVERNMENT REPRESENTATIVE:

SIGNATURE _____ DATE _____

FIGURE 4. Sample inspection record.

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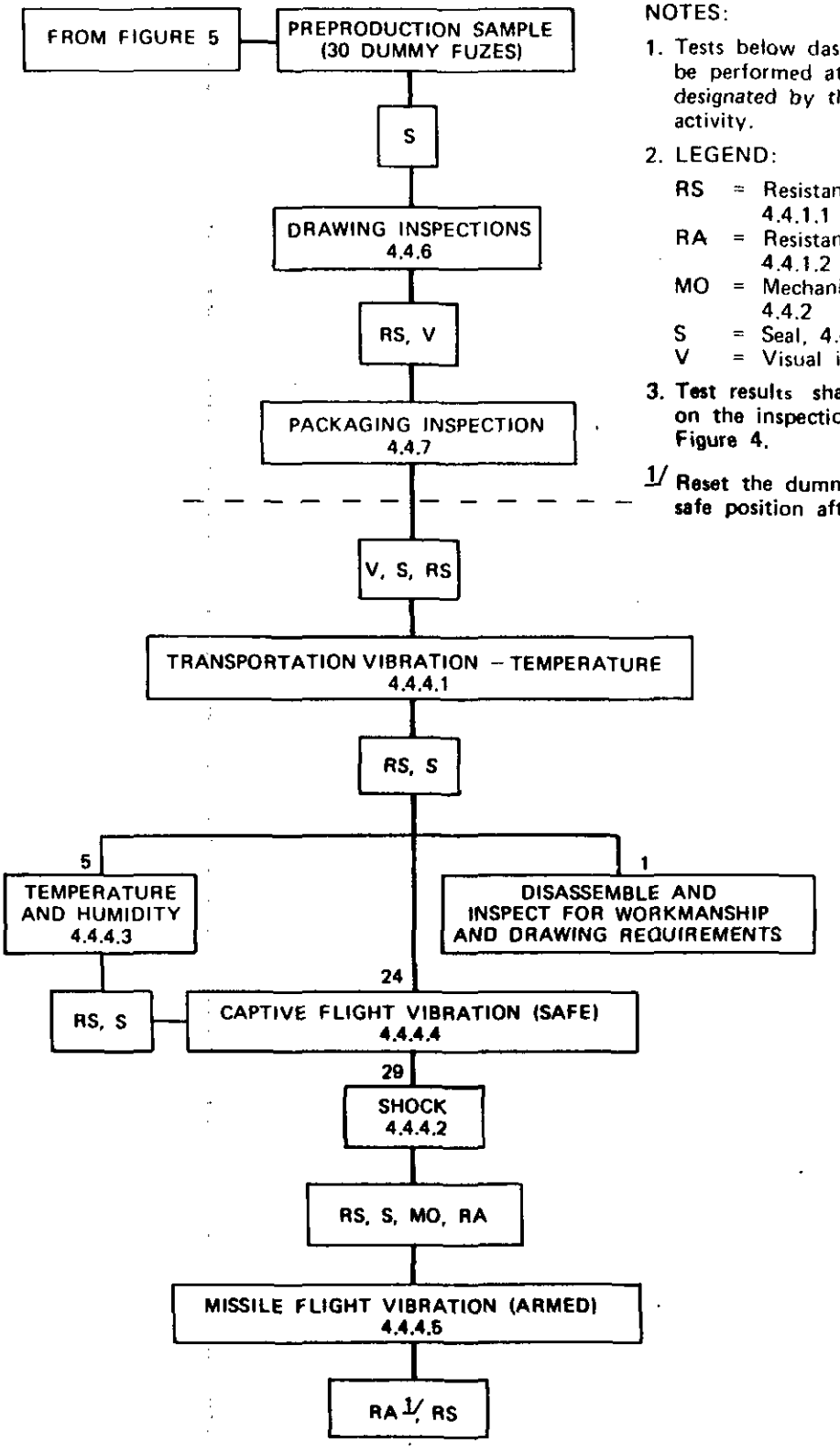
NOTES:

1. Test results shall be recorded on the inspection record of Figure 4.

1/ Reset the dummy fuze to the safe position after test (see 4.3.5).

FIGURE 5. Individual unit acceptance inspection sequence.

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NOTES:

1. Tests below dashed line shall be performed at a facility designated by the procuring activity.
2. LEGEND:
   - RS = Resistance (safe position), 4.4.1.1
   - RA = Resistance (armed position), 4.4.1.2
   - MO = Mechanical operability, 4.4.2
   - S = Seal, 4.4.3
   - V = Visual inspection, 4.4.5
3. Test results shall be recorded on the inspection record of Figure 4.

1/ Reset the dummy fuze to the safe position after test (see 4.3.5).

FIGURE 6. Preproduction sample inspection sequence.

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NOTES:

1. Tests below dashed line shall be performed at a facility designated by the procuring activity.
2. LEGEND:
   - RS = Resistance (safe position), 4.4.1.1
   - RA = Resistance (armed position), 4.4.1.2
   - MO = Mechanical operability, 4.4.2
   - S = Seal, 4.4.3
   - V = Visual inspection, 4.4.5
3. Test results shall be recorded on the inspection record of Figure 4.

1/ Reset the dummy fuze to the safe position after test (see 4.3.5).

2/ Remainder of lot shall be held in bond pending results of tests on 15 sample dummy fuzes. If sample dummy fuzes do not cause lot rejection, the remainder of lot shall continue through indicated test sequence.

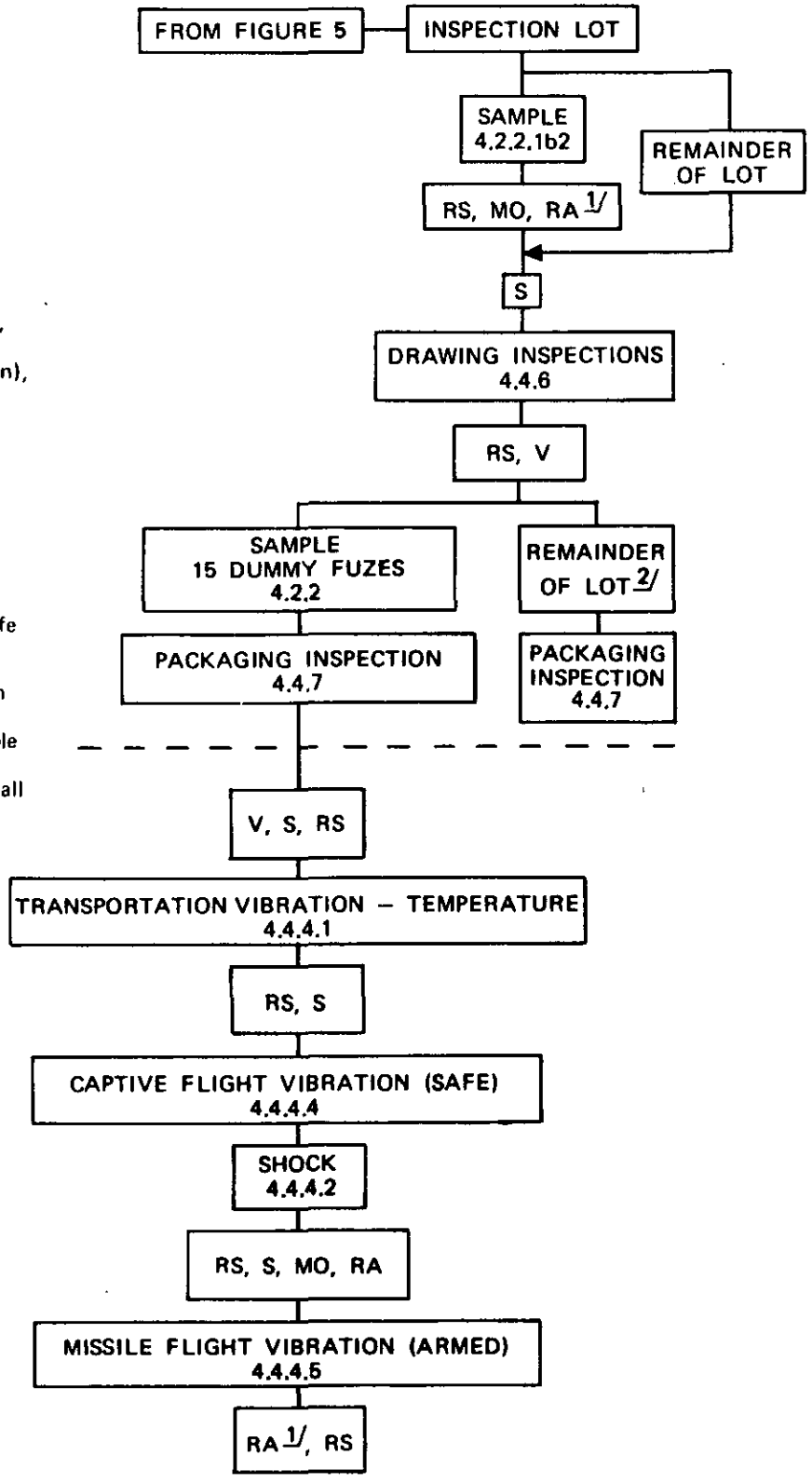


FIGURE 7. Inspection lot inspection sequence.

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-F-85400(AS), Fuze, Guided Missile, Dummy, FMU-138(D-1)/B

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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